intro

themes

automating buidling software

- libraries, taking advantage of incremental compilation

sharing machines

- multiple users/programs on one system

parallelism and concurrency

- doing two+ things at once

communicating between machines

- layered design of networks
- implementing secure communication

themes

automating buidling software

libraries, taking advantage of incremental compilation

sharing machines

multiple users/programs on one system

parallelism and concurrency

doing two+ things at once

communicating between machines

layered design of networks

implementing secure communication

make

```
$ ./foo.exe
...
...
$ edit readline.c
$ make
clang -g -O -Wall -c readline.c -o readline.o
ar rcs libreadline.a terminal.o readline.o
clang -o foo.exe foo.o foo-utility.o -L. -lreadline
$
```

themes

automating buidling software

- libraries, taking advantage of incremental compilation

sharing machines

- multiple users/programs on one system

parallelism and concurrency

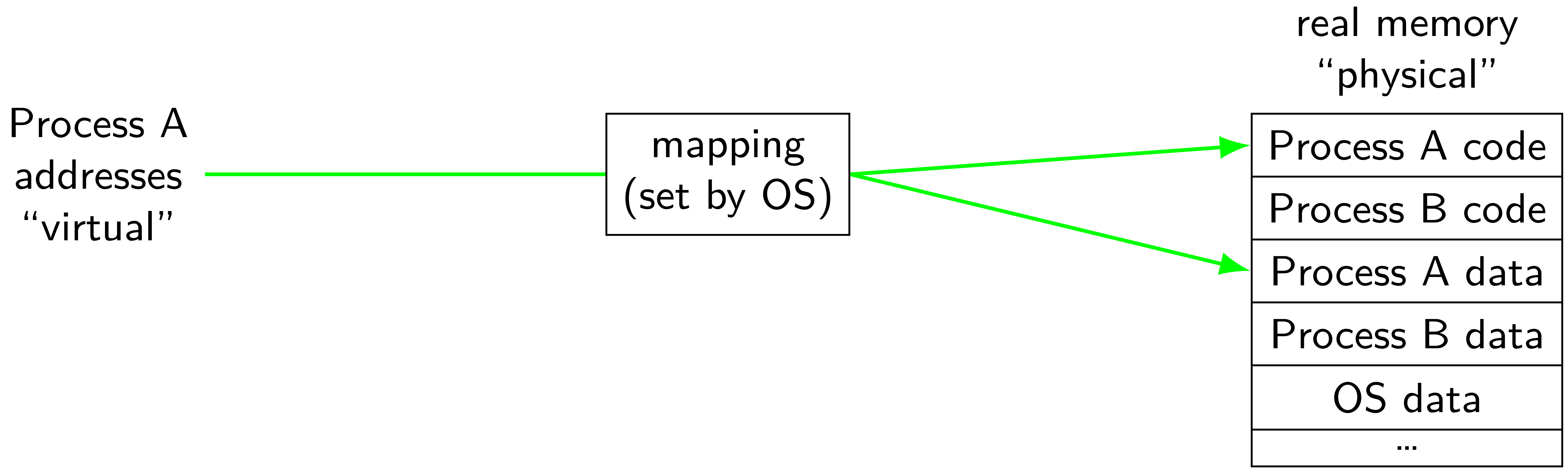
- doing two+ things at once

communicating between machines

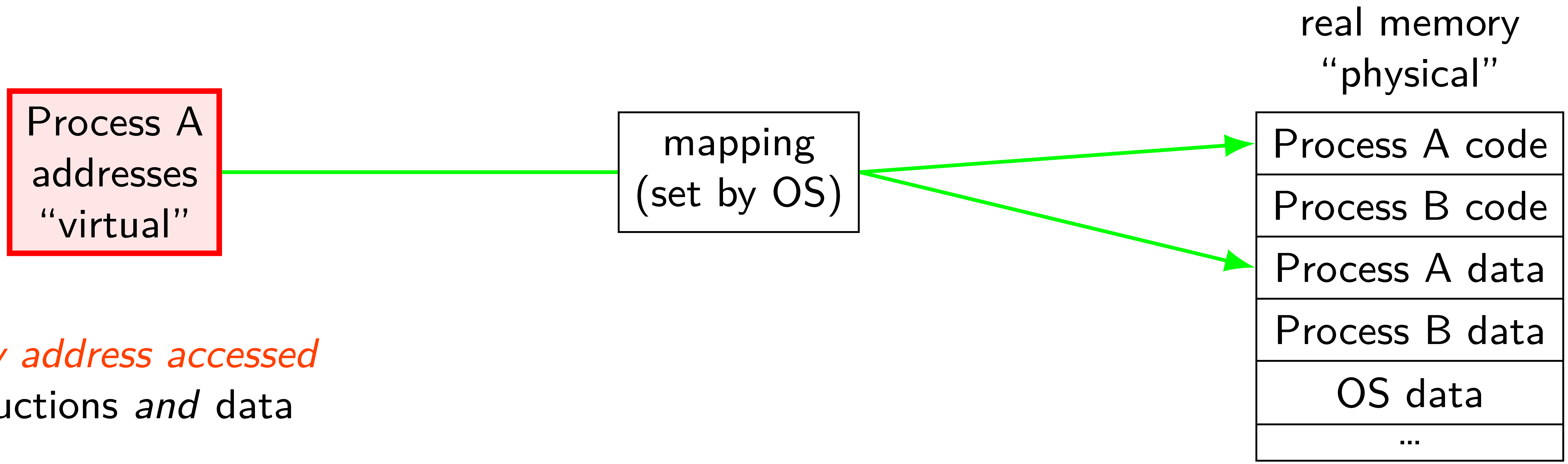
- layered design of networks
- implementing secure communication

address translation

address translation

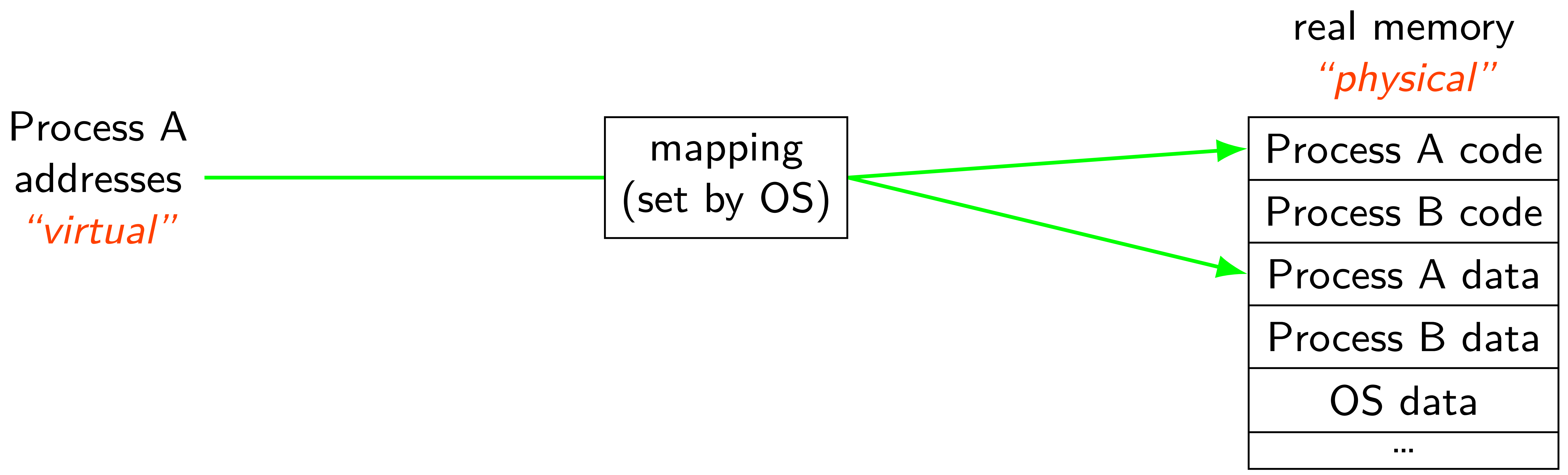


address translation



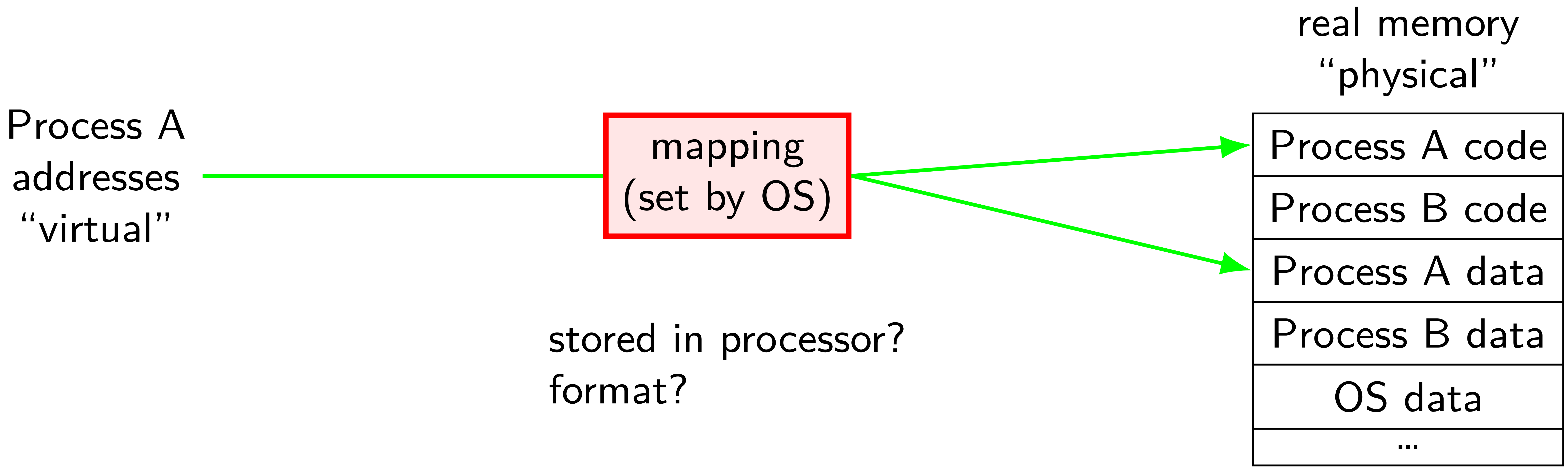
every address accessed
instructions *and* data

address translation



program addresses are 'virtual'
real addresses are 'physical'
can be *different sizes*!

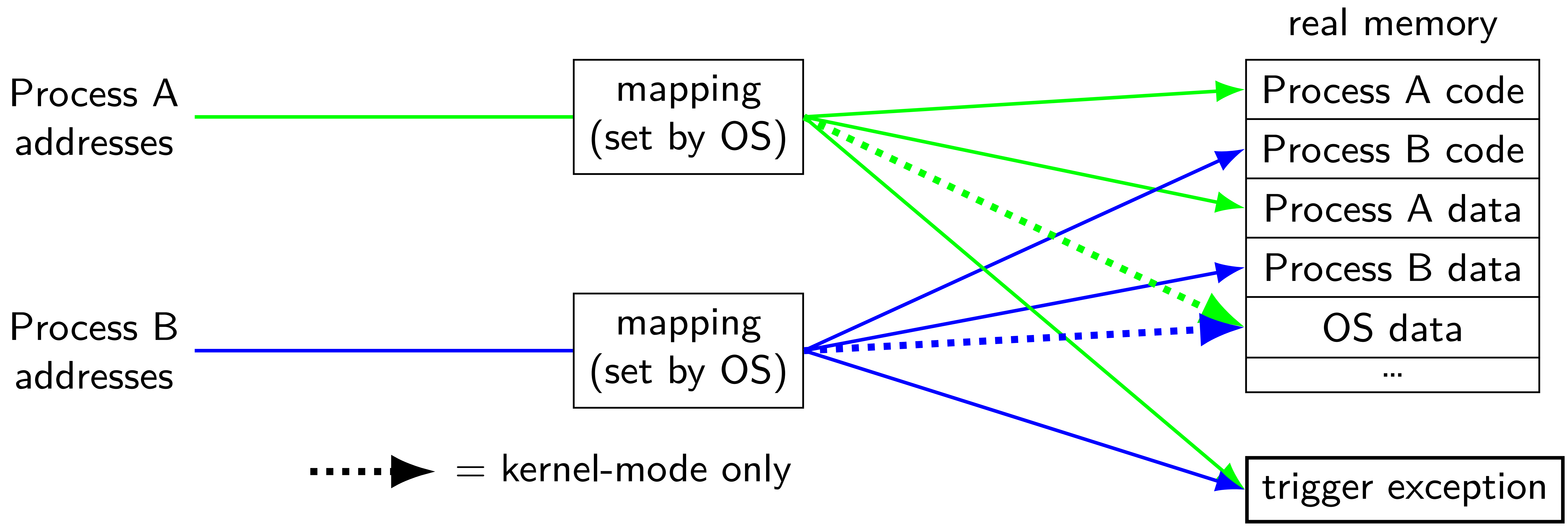
address translation



stored in processor?
format?

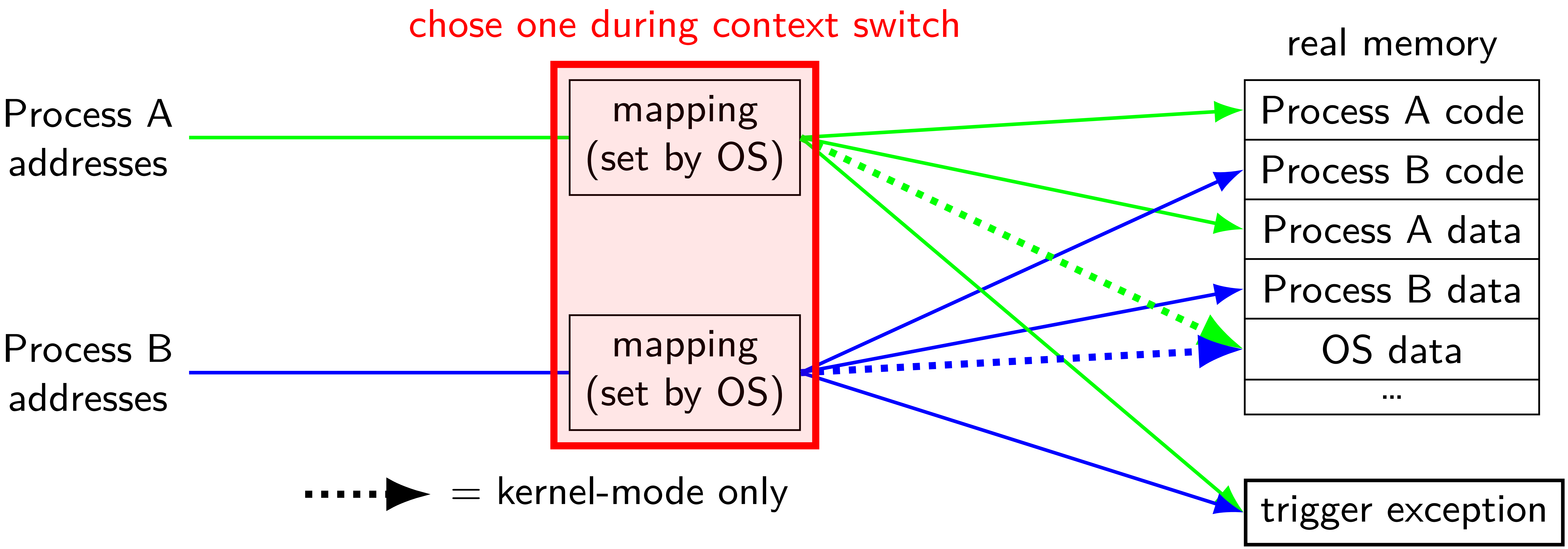
address spaces

illusion of *dedicated memory*



address spaces

illusion of *dedicated memory*



themes

automating buidling software

libraries, taking advantage of incremental compilation

sharing machines

multiple users/programs on one system

parallelism and concurrency

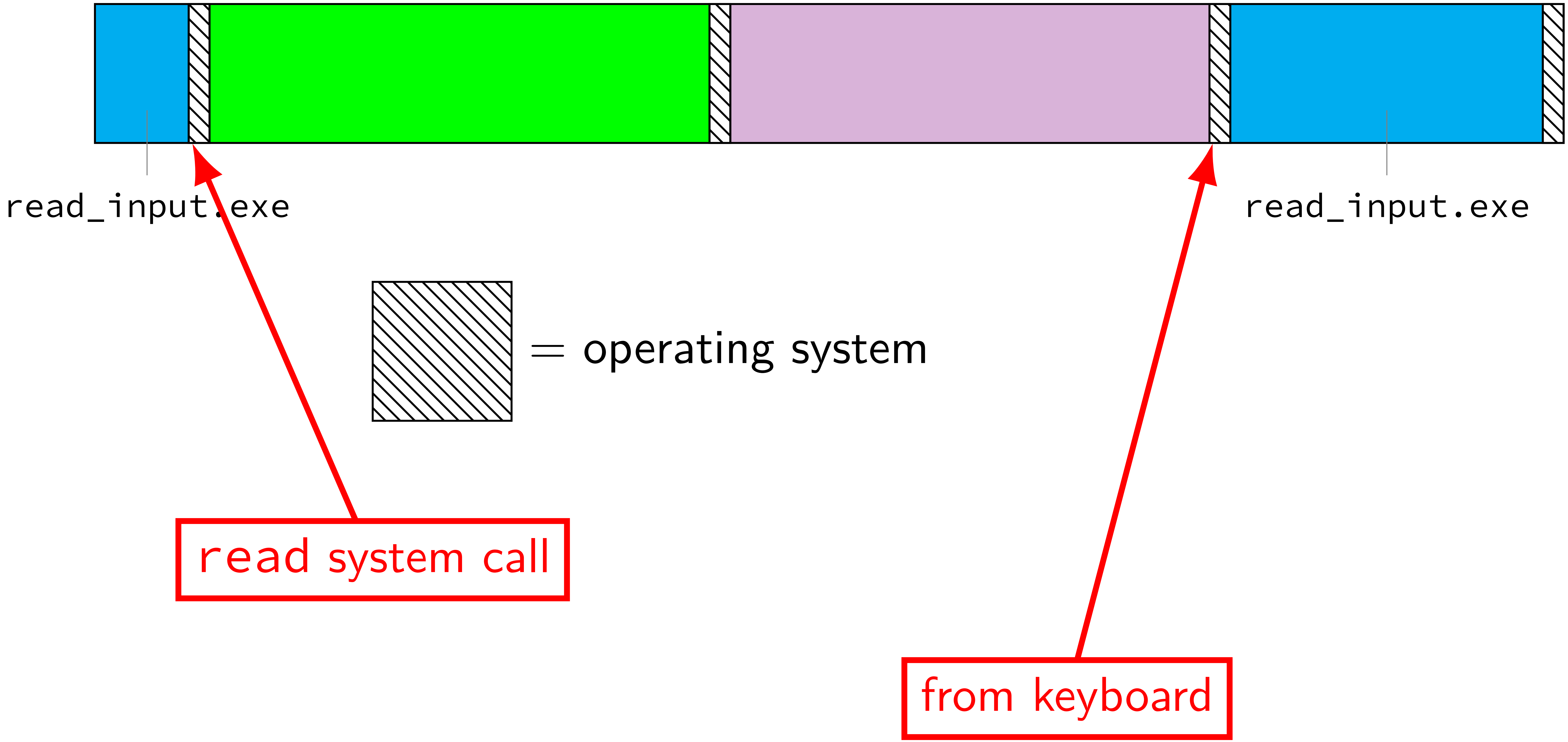
doing two+ things at once

communicating between machines

layered design of networks

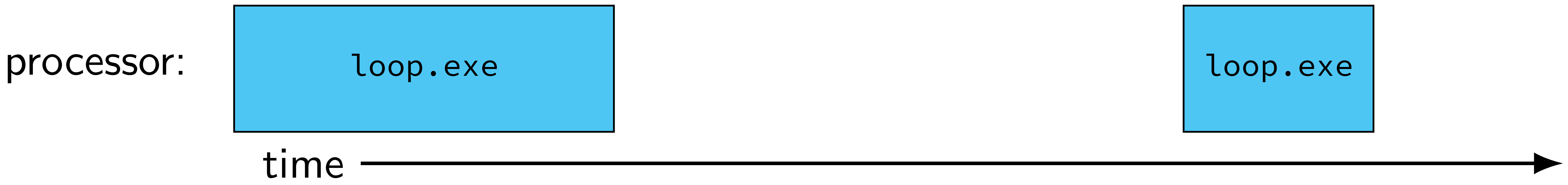
implementing secure communication

keyboard input timeline



time multiplexing

time multiplexing



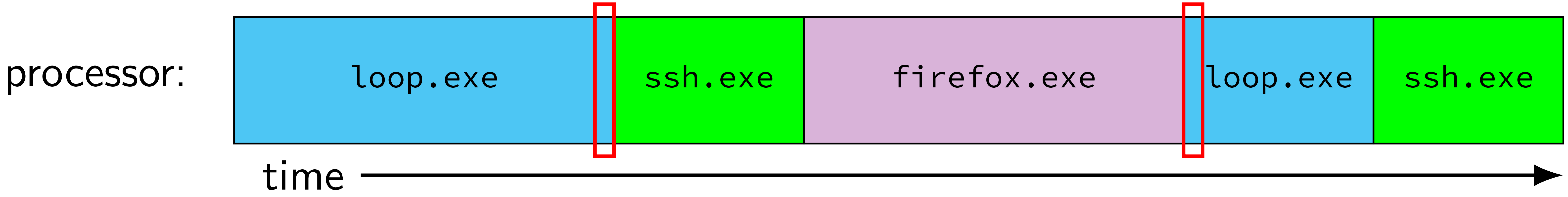
time multiplexing


```
...
loop: ...
      ...
      jmp loop
loop: ...
      ...
```

million cycle delay

```
      ...
      jmp loop
loop: ...
      ...
```

time multiplexing

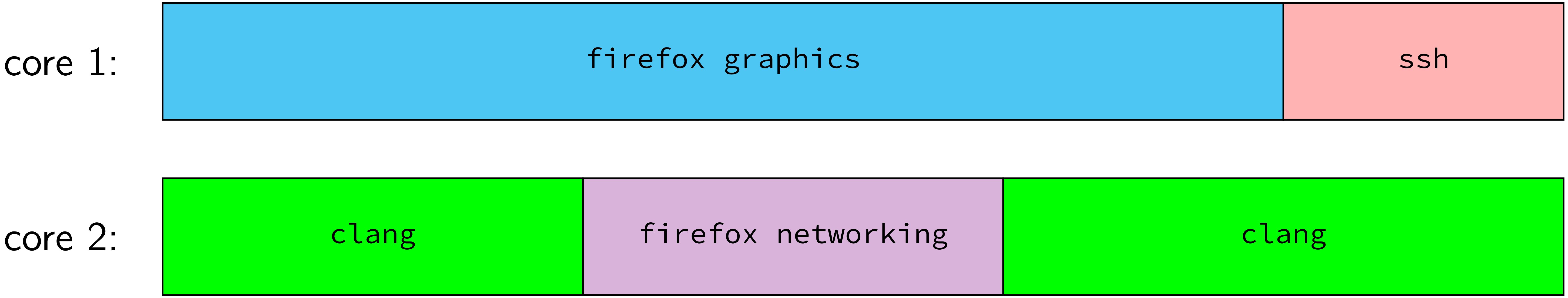



```
...
loop: ...
      ...
      jmp loop
loop: ...
      ...
```

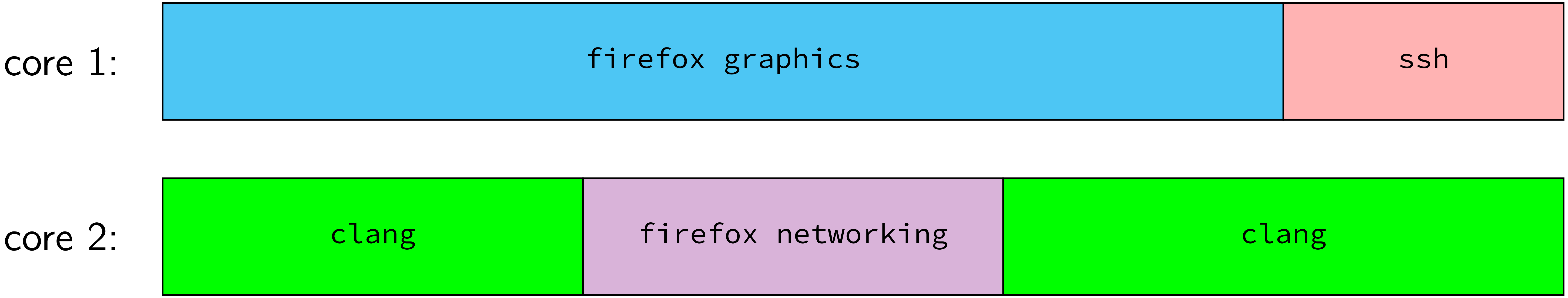
million cycle delay

```
      ...
      jmp loop
loop: ...
      ...
```

multiple cores+threads

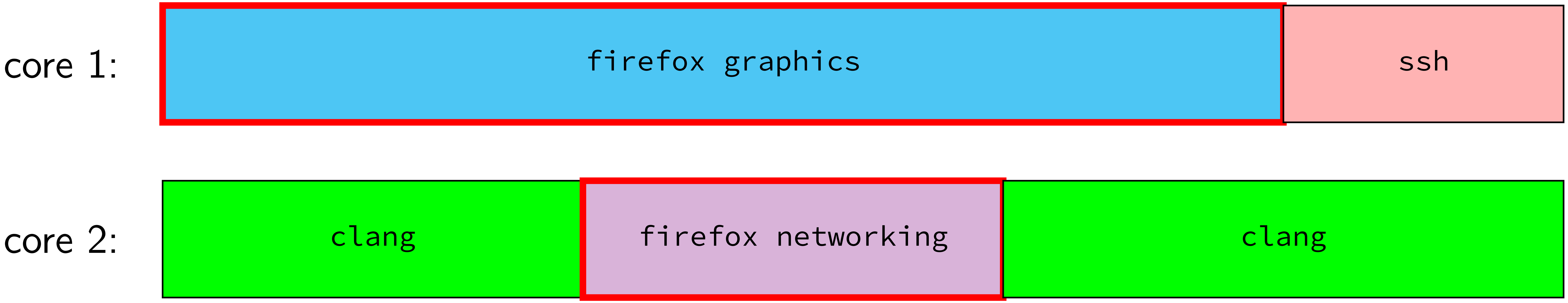


multiple cores+threads



multiple cores? each core still divided up

multiple cores+threads



multiple cores? each core still divided up

one program with multiple threads

permissions

```
$ ls /u/other/secret
ls: cannot open directory '/u/other/secret': Permission denied
$ shutdown
shutdown: Permission denied
```

themes

automating buidling software

- libraries, taking advantage of incremental compilation

sharing machines

- multiple users/programs on one system

parallelism and concurrency

- doing two+ things at once

communicating between machines

- layered design of networks
- implementing secure communication

networking layers

application	HTTP, SSH, SMTP, …	application-defined meanings
transport	TCP, UDP, …	reach correct program reliability/streams
network	IPv4, IPv6	reach correct machine (across networks)
link	Ethernet, Wi-Fi, …	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, …	encode bits for wire/radio

networking layers

application	HTTP, SSH, SMTP, …	application-defined meanings
transport	TCP, UDP, …	reach correct program *reliability/streams*
network	IPv4, IPv6	reach correct machine (across networks)
link	Ethernet, Wi-Fi, …	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, …	encode bits for wire/radio

networking layers

application	HTTP, SSH, SMTP, …	application-defined meanings
transport	TCP, UDP, …	reach correct program reliability/streams
network	IPv4, IPv6	reach correct machine (across networks)
link	Ethernet, Wi-Fi, …	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, …	encode bits for wire/radio

networking layers

application	HTTP, SSH, SMTP, …	application-defined meanings
transport	TCP, UDP, …	reach correct program reliability/streams
network	IPv4, IPv6	reach correct machine (across networks)
link	Ethernet, Wi-Fi, …	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, …	encode bits for wire/radio

networking layers

application	HTTP, SSH, SMTP, …	application-defined meanings
transport	TCP, UDP, …	reach correct program reliability/streams
network	IPv4, IPv6	reach correct machine (across networks)
link	Ethernet, Wi-Fi, …	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, …	encode bits for wire/radio

networking layers

application	HTTP, SSH, SMTP, …	application-defined meanings
transport	TCP, UDP, …	reach correct program reliability/streams
network	IPv4, IPv6	reach correct machine (across networks)
link	Ethernet, Wi-Fi, …	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, …	encode bits for wire/radio

layers terminology

application	application-defined meanings	
transport	reach correct program, reliablity/streams	segments/datagrams
network	reach correct machine (across networks)	packets
link	coordinate shared wire/radio	frames
physical	encode bits for wire/radio	

names and addresses

name	*address*
logical identifier	*location/how to locate*
variable `counter`	memory address `0x7FFF9430`
DNS name `www.virginia.edu`	IPv4 address `128.143.22.36`
DNS name `mail.google.com`	IPv4 address `216.58.217.69`
DNS name `mail.google.com`	IPv6 address `2607:f8b0:4004:80b::2005`
DNS name `cso2.cs.virginia.edu`	IPv4 address `128.143.67.91`
DNS name `cso2.cs.virginia.edu`	MAC address `18:66:da:2e:7f:da`
service name `https`	port number `443`
service name `ssh`	port number `22`

secure communication?

how do you know who your socket is to?

who can read what's on the socket?

what can you do to restrict this?

themes

automating buidling software

- libraries, taking advantage of incremental compilation

sharing machines

- multiple users/programs on one system

parallelism and concurrency

- doing two+ things at once

communicating between machines

- layered design of networks
- implementing secure communication

2004 CPU

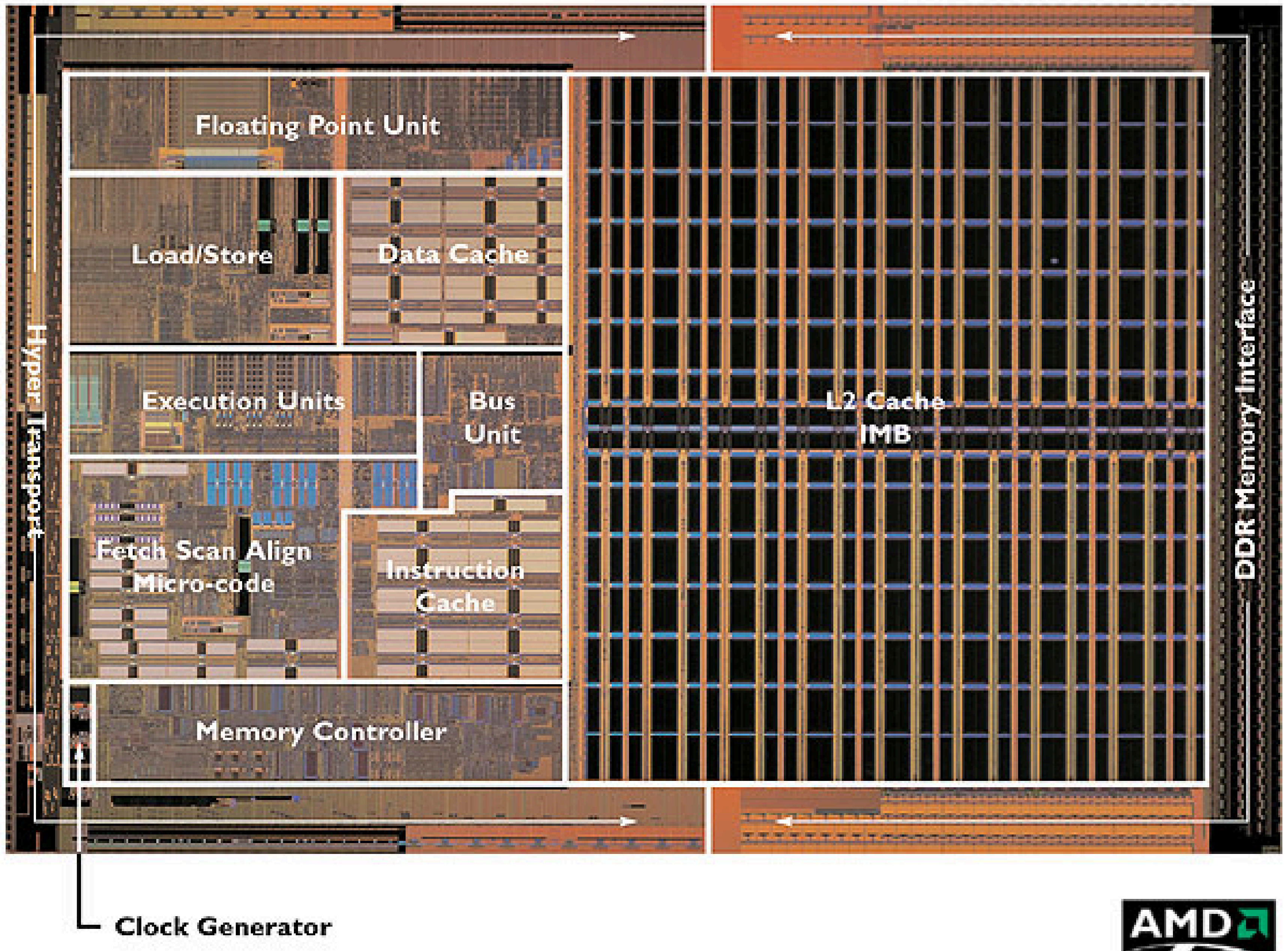


Image: approx 2004 AMD press image of Opteron die;
approx register location via chip-architect.org (Hans de Vries

2004 CPU

Registers

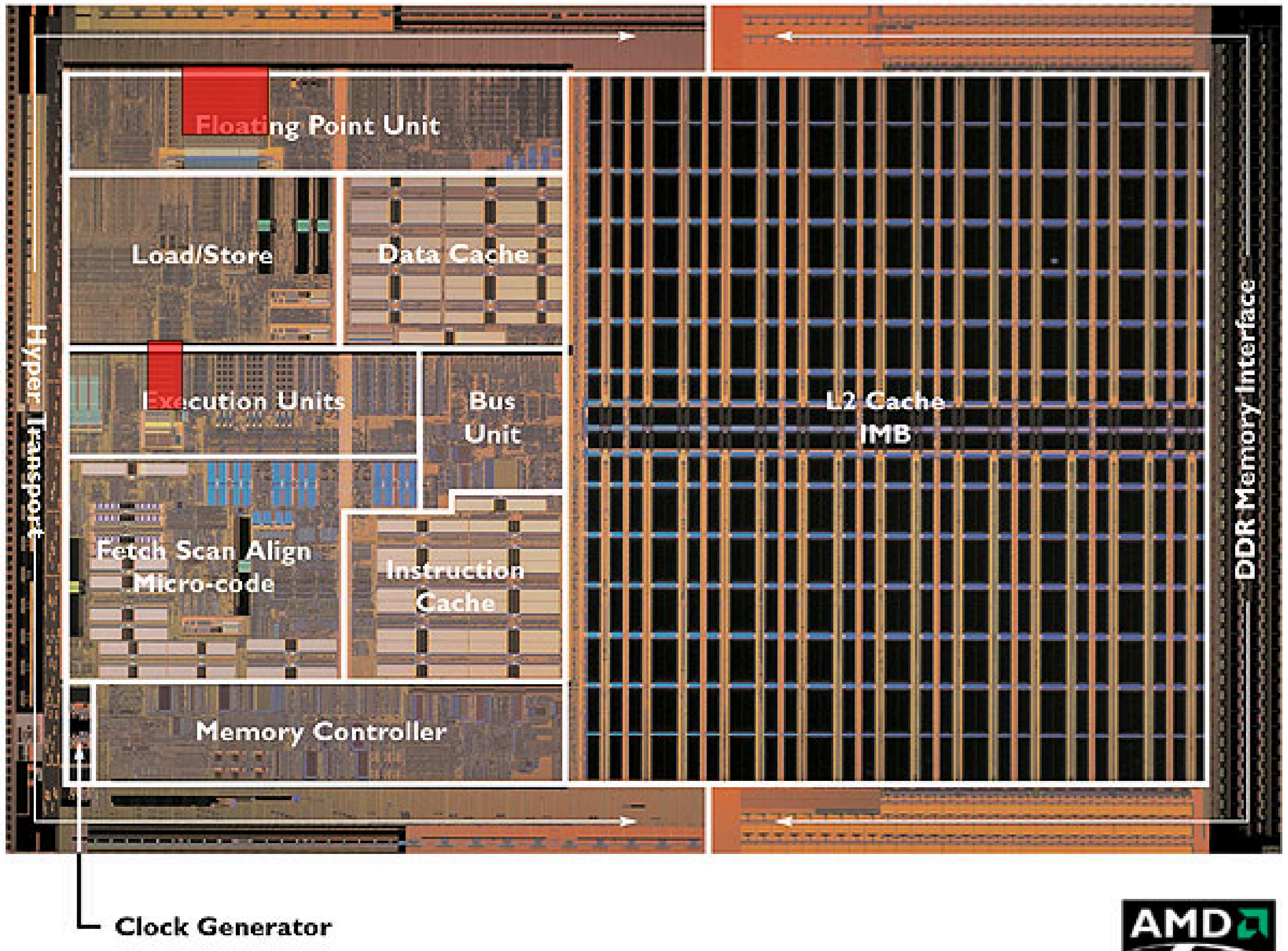
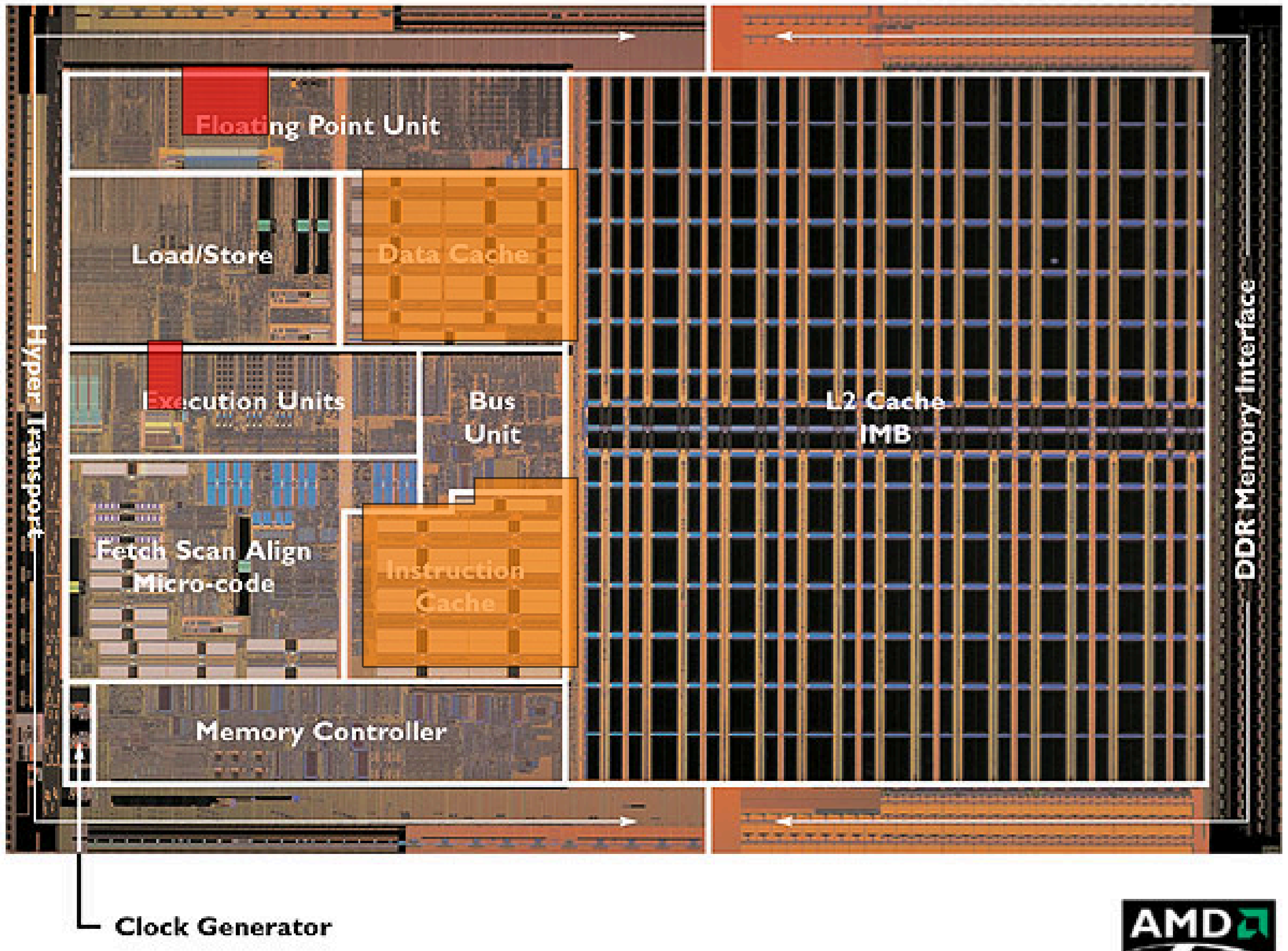


Image: approx 2004 AMD press image of Opteron die;
approx register location via chip-architect.org (Hans de Vries

2004 CPU



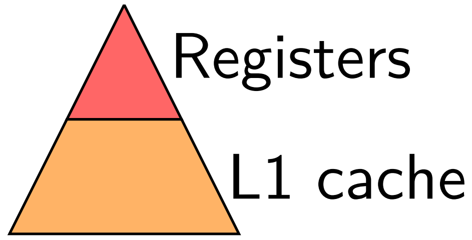
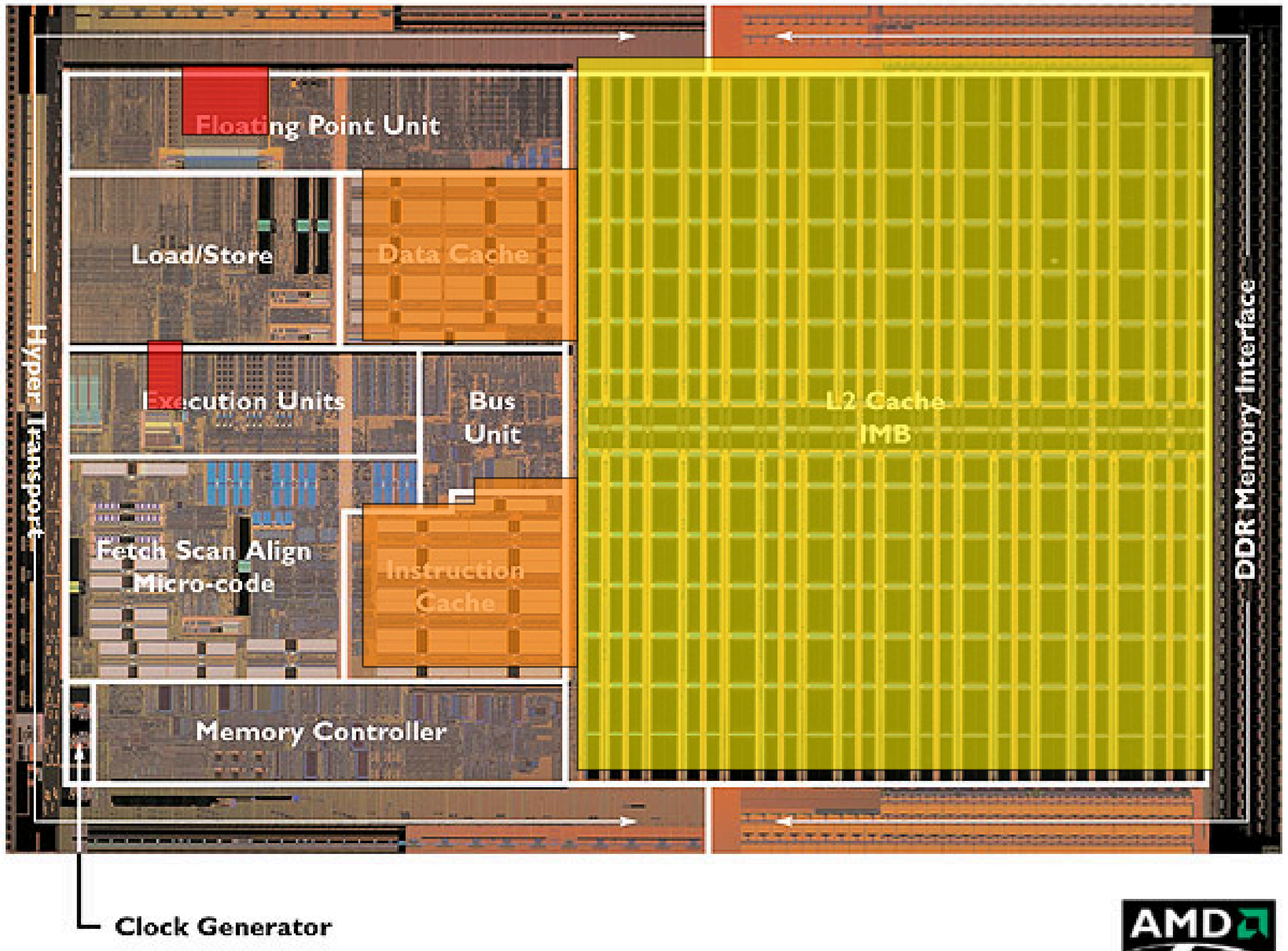


Image: approx 2004 AMD press image of Opteron die;
approx register location via chip-architect.org (Hans de Vries)

2004 CPU



AMD 64 Opteron

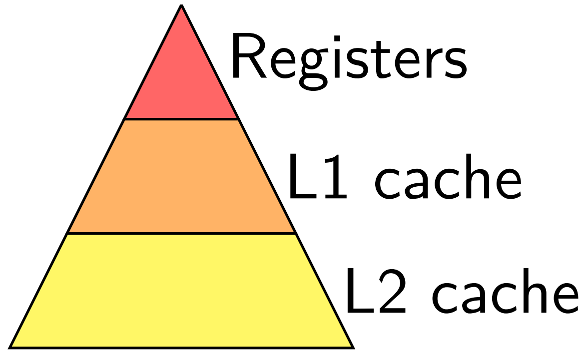
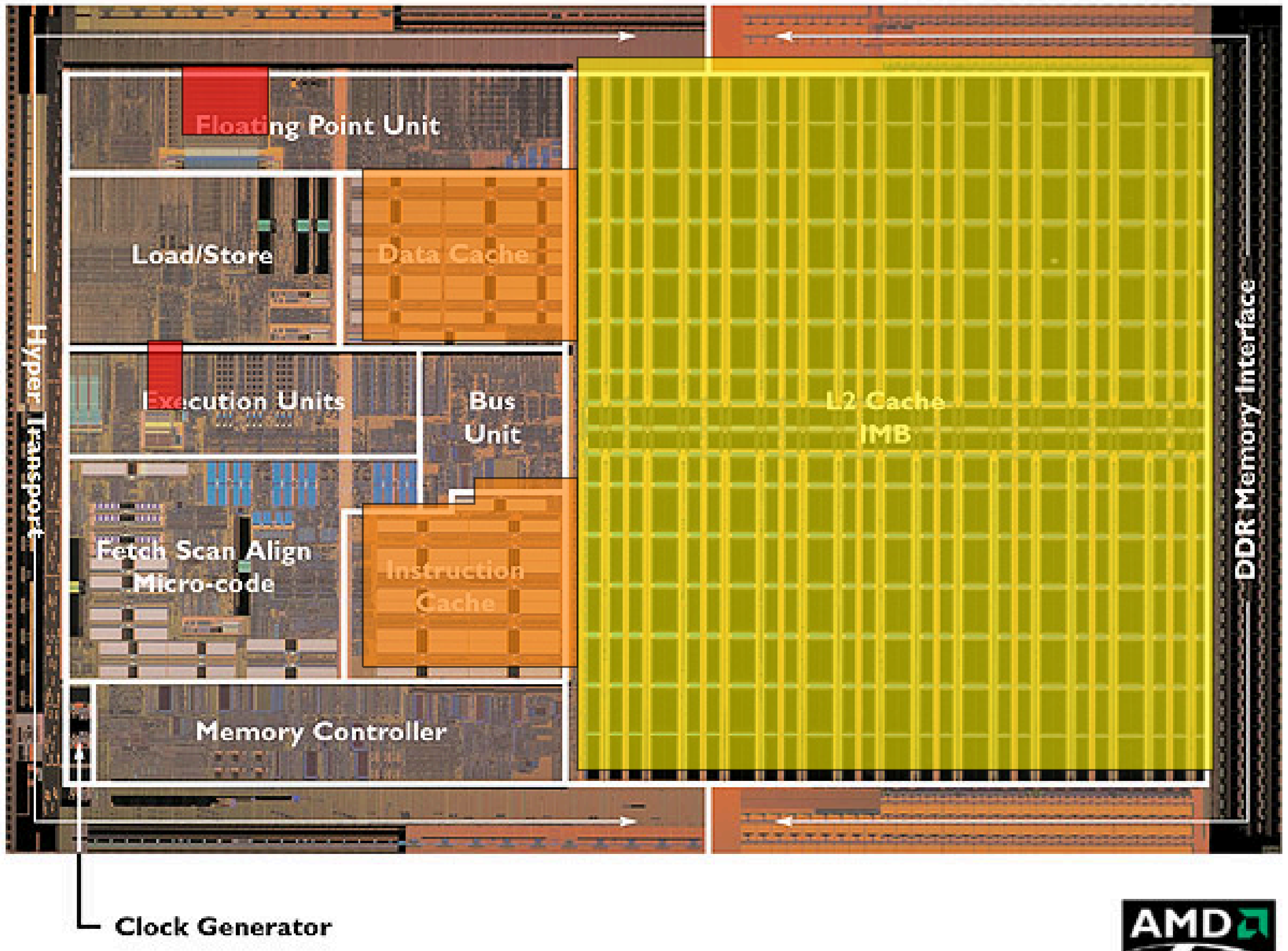


Image: approx 2004 AMD press image of Opteron die;
approx register location via chip-architect.org (Hans de Vries

2004 CPU



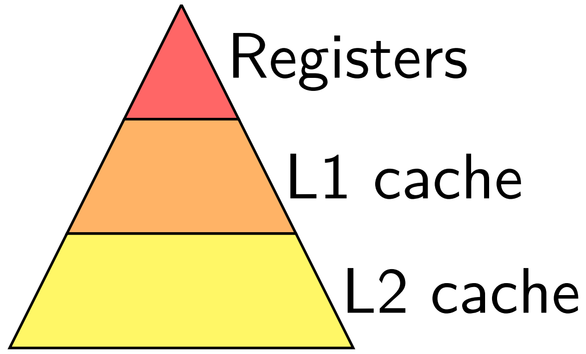
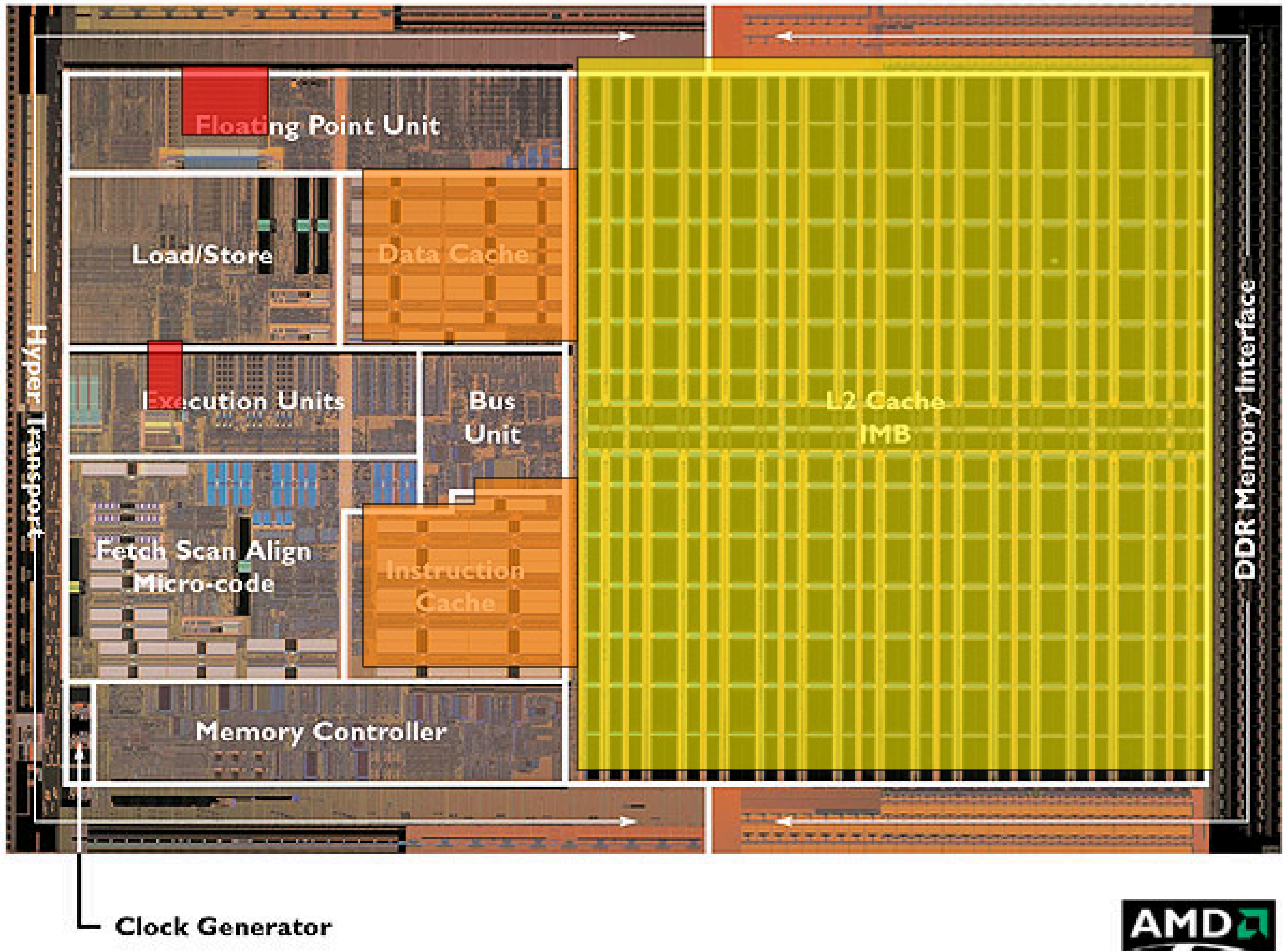


Image: approx 2004 AMD press image of Opteron die;
approx register location via chip-architect.org (Hans de Vries

2004 CPU



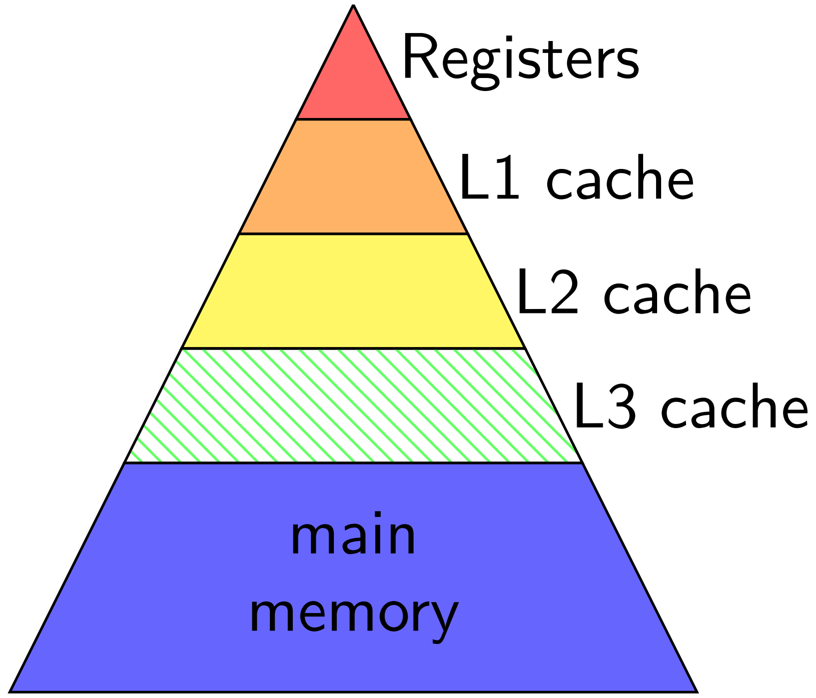
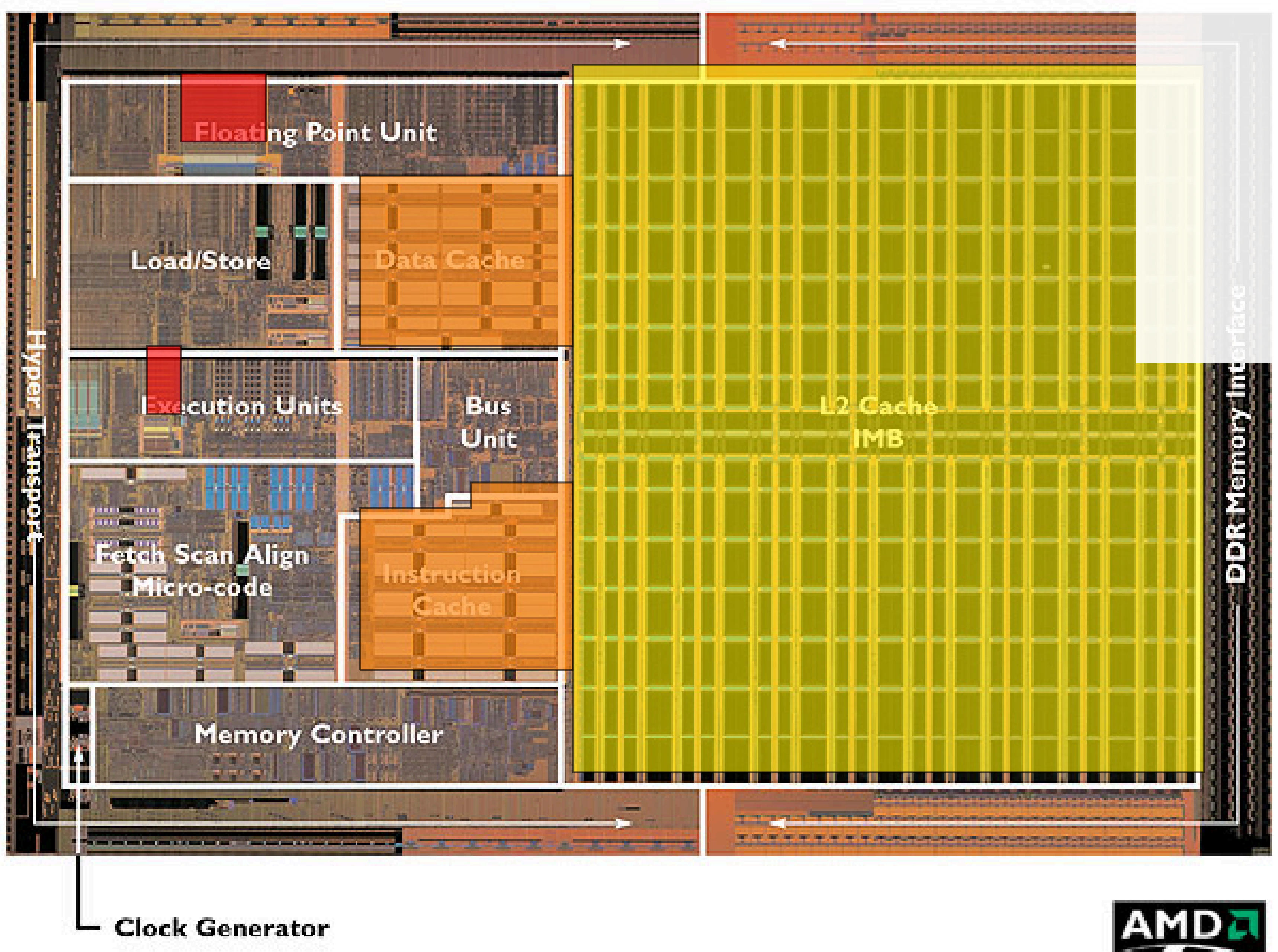


Image: approx 2004 AMD press image of Opteron die;
approx register location via chip-architect.org (Hans de Vries

2004 CPU



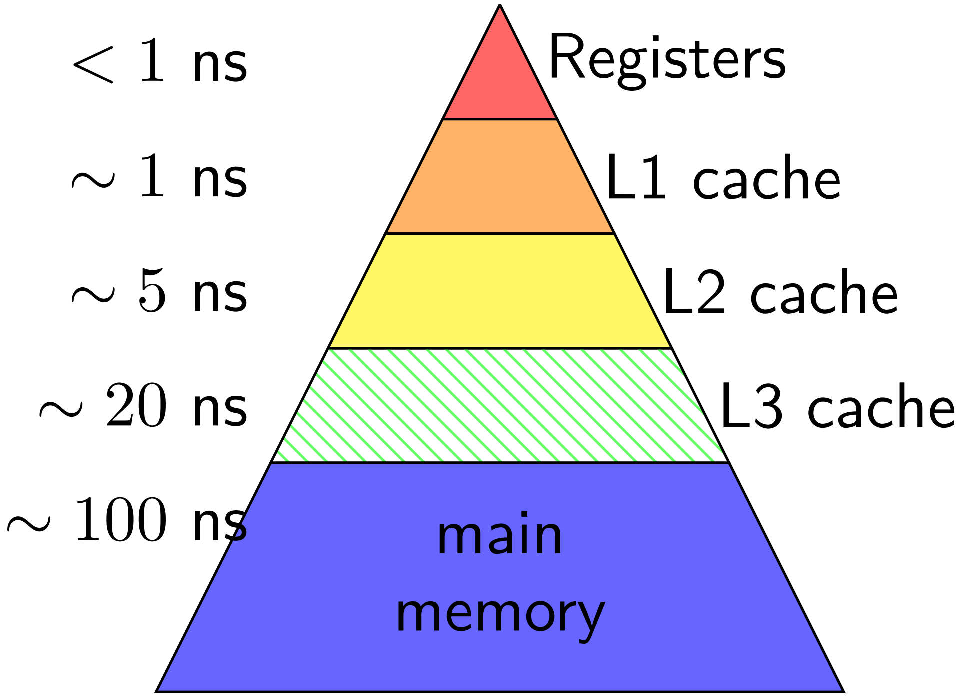


Image: approx 2004 AMD press image of Opteron die;
approx register location via chip-architect.org (Hans de Vries

some performance examples

some performance examples

```
example1:
    movq $10000000000, %rax
loop1:
    addq %rbx, %rcx
    decq %rax
    jge loop1
    ret
```

about 30B instructions

my desktop: approx 2.65 sec

```
example2:
    movq $10000000000, %rax
loop2:
    addq %rbx, %rcx
    addq %r8, %r9
    decq %rax
    jge loop2
    ret
```

about 40B instructions

my desktop: approx 2.65 sec

some performance examples

```
example1:
    movq $10000000000, %rax
loop1:
    addq %rbx, %rcx
    decq %rax
    jge loop1
    ret
```

about 30B instructions

my desktop: approx *2.65 sec*

```
example2:
    movq $10000000000, %rax
loop2:
    addq %rbx, %rcx
    addq %r8, %r9
    decq %rax
    jge loop2
    ret
```

about 40B instructions

my desktop: approx *2.65 sec*

some performance examples

```
example1:
  movq $10000000000, %rax
loop1:
  addq %rbx, %rcx
  decq %rax
  jge loop1
  ret
```

about *30B instructions*

my desktop: *2.65 sec*

```
example1:
  movq $10000000000, %rax
loop1:
  addq %rbx, %rcx
  addq %r8,  %r9
  decq %rax
  jge loop1
  ret
```

about *40B instructions*

my desktop: *2.65 sec*

C exercise

```
int array[4] = {10,20,30,40};
int *p;
p = &array[0];
p += 2;
p[1] += 1;
```

array =

A. compile or runtime error
B. {10,20,30,41}
C. {10,20,32,41}
D. {10,21,30,40}
E. {12,21,30,40}
F. none of these

C exercise (2)

```
int *array2[4]; int array1[4] = {10,20,30,40};
void mystery(int **p) {
    *p = &array1[2];
}
int main() {
    int **q;
    q = array2;
    mystery(q);
    array1[1] = *q;
    ...
}
```

array1 =

A. compile or runtime error
B. {10,10,30,40}
C. {10,30,30,40}
D. {10,10,20,30}
E. {10,20,10,20}
F. none of these

C exercise (2)

```
int *array2[4]; int array1[4] = {10,20,30,40};
void mystery(int **p) {
    *p = &array1[2];
}
int main() {
    int **q;
    q = array2;
    mystery(q);
    array1[1] = *q;
    ...
}
```

array1 =

A. compile or runtime error
B. {10,10,30,40}
C. {10,30,30,40}
D. {10,10,20,30}
E. {10,20,10,20}
F. none of these

some avenues for review

review CSO1 stuff; examples from Fall 2025:

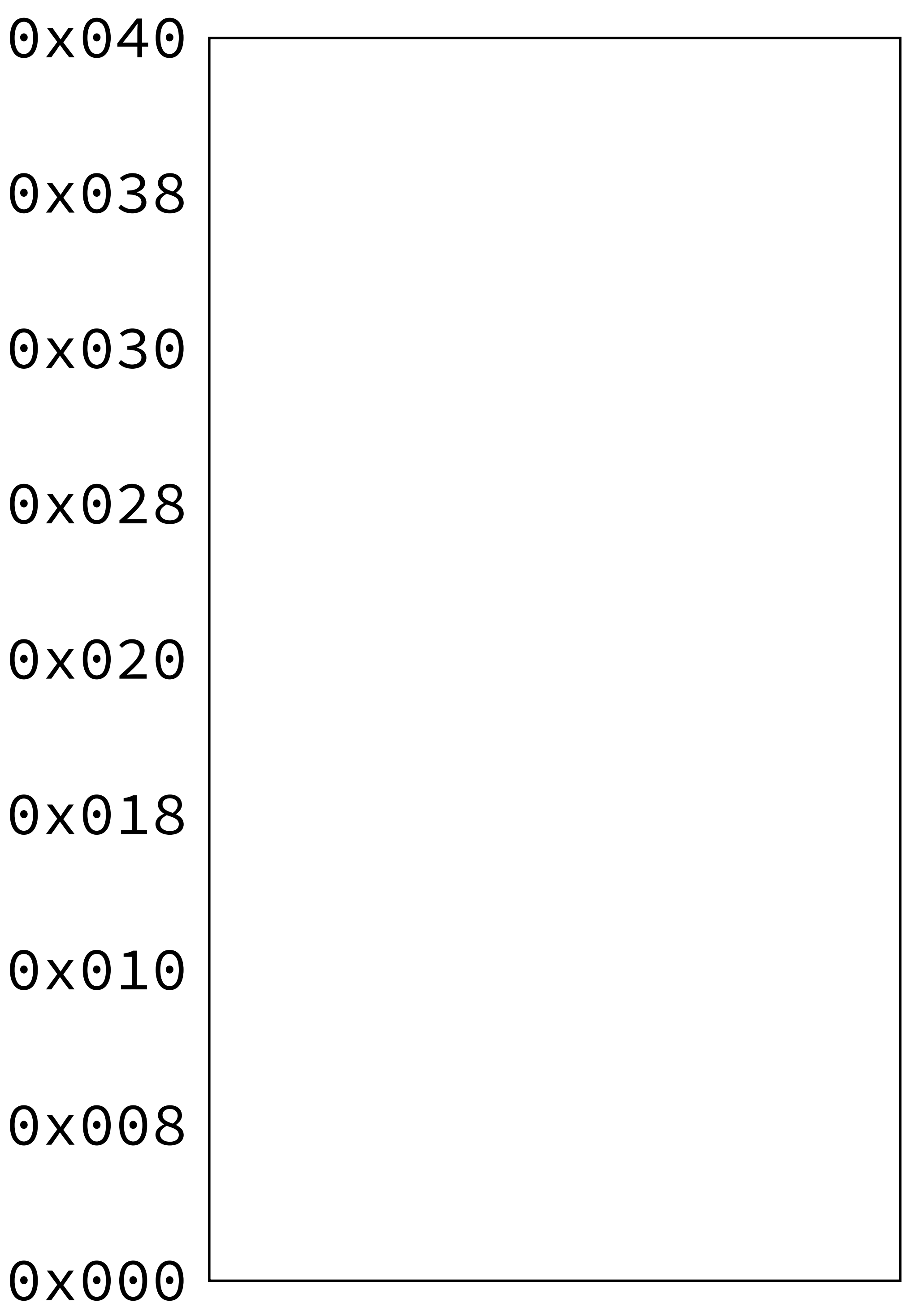
- labs 10–13, homeworks 7–8
- https://researcher111.github.io/uva-cso1-F23-DG/

exercises we've used in the past:

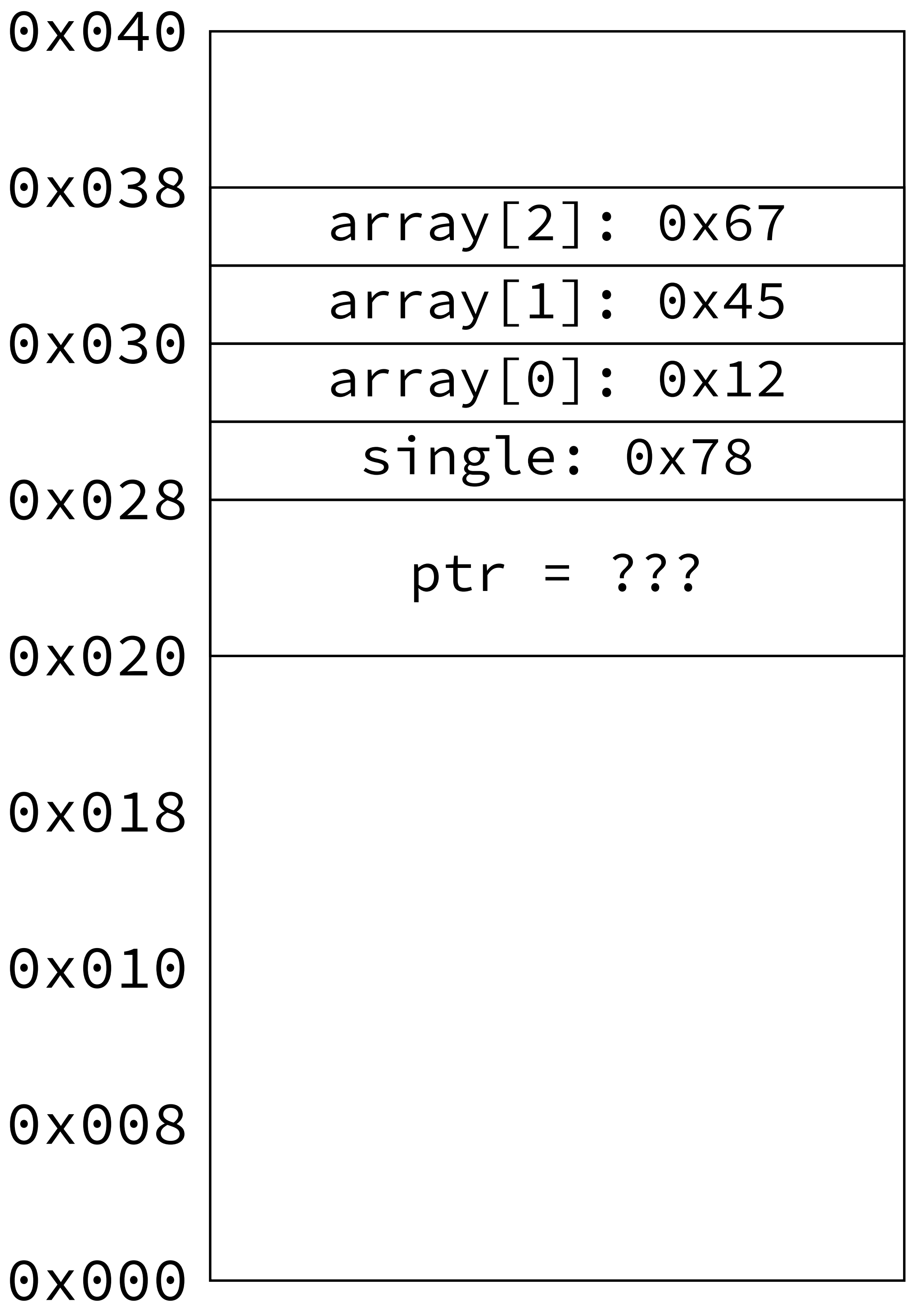
- implement strsep library function
- implement conversion from dynamic array to linked list

some pointer stuff

```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;
```

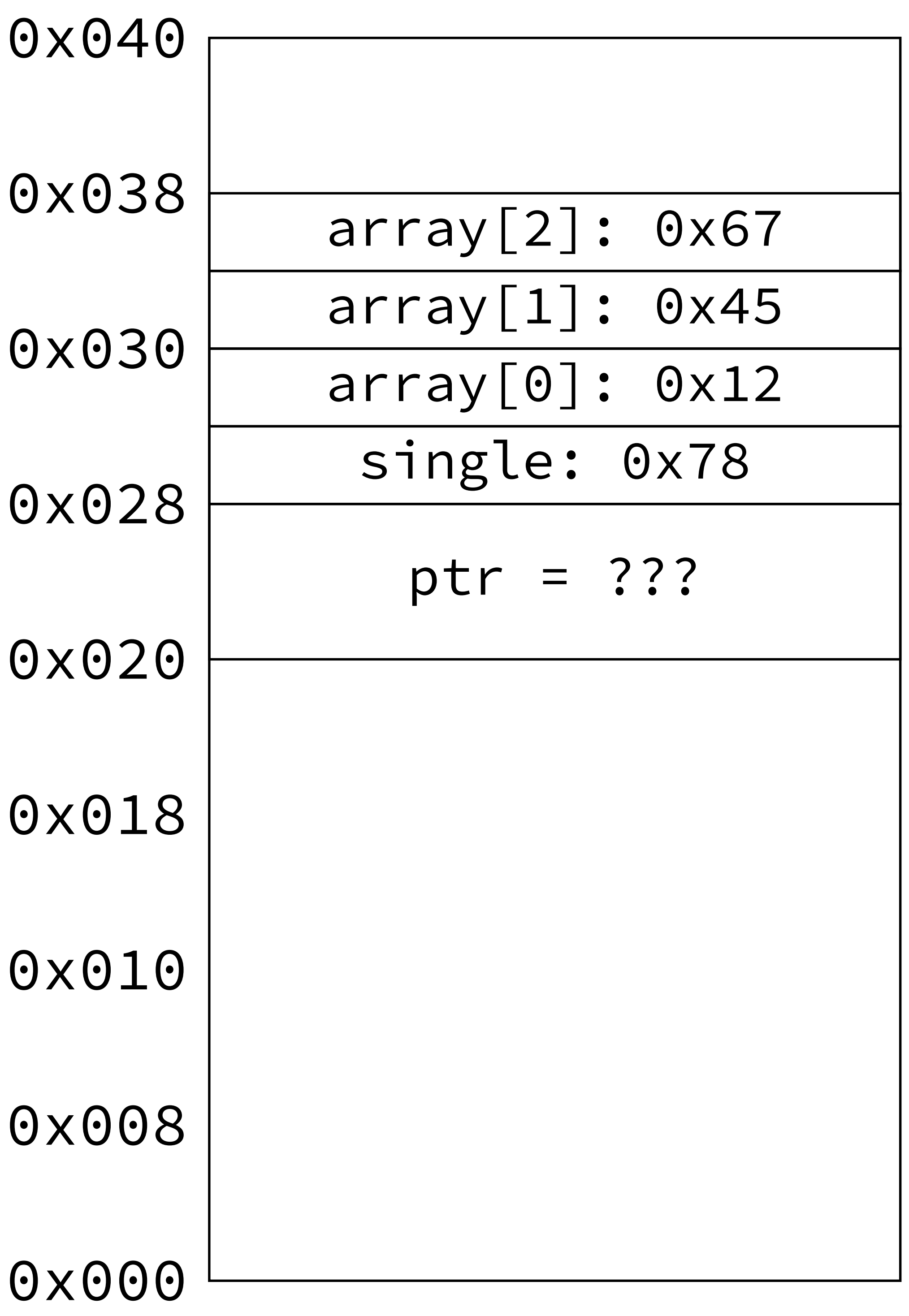


some pointer stuff




```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;
```

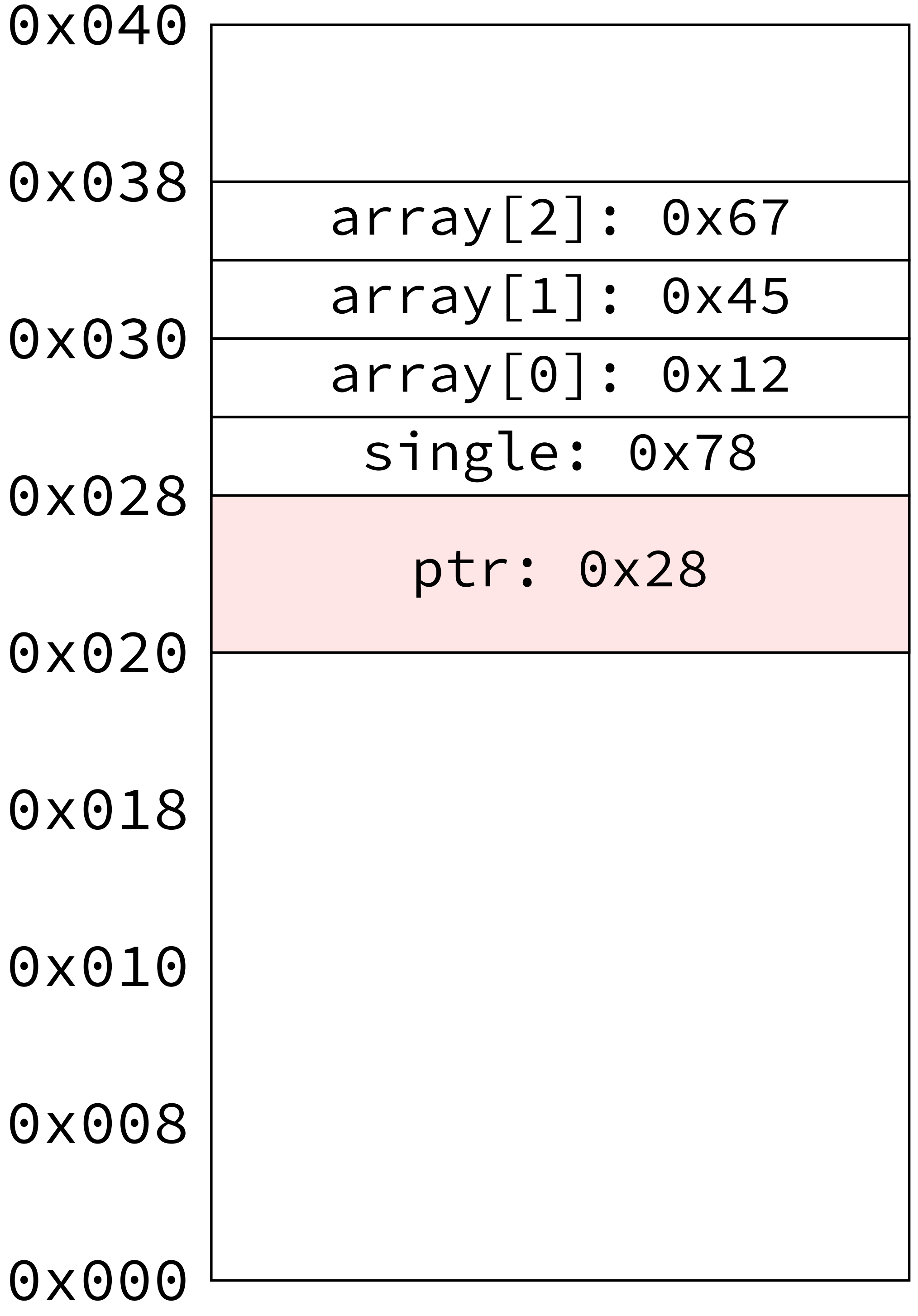
some pointer stuff




```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;
```

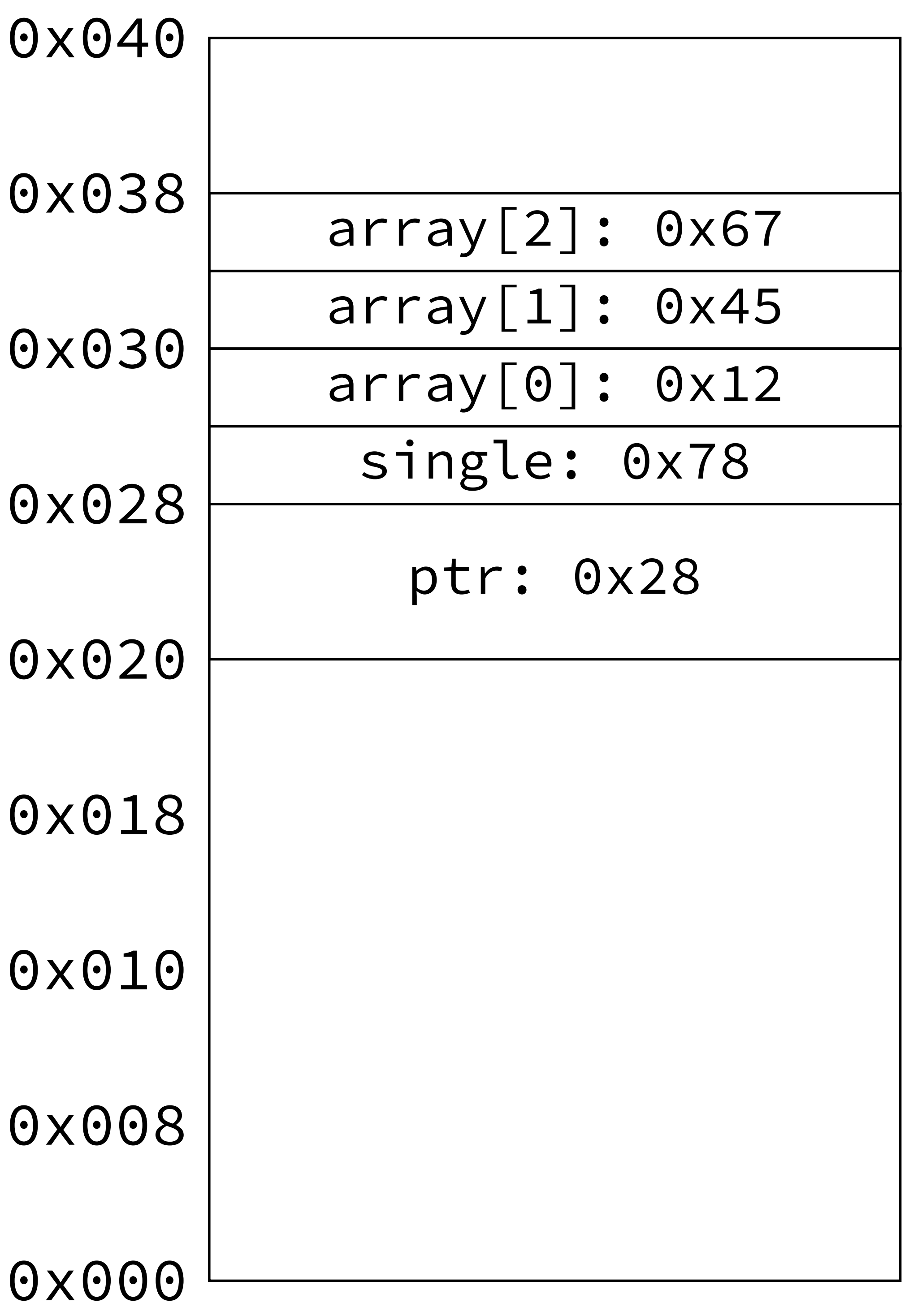
~~`*ptr = 0xAB;`~~ runtime error

some pointer stuff




```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;

ptr = &single;
ptr = (int*) 0x28;
```
addr. of single

some pointer stuff



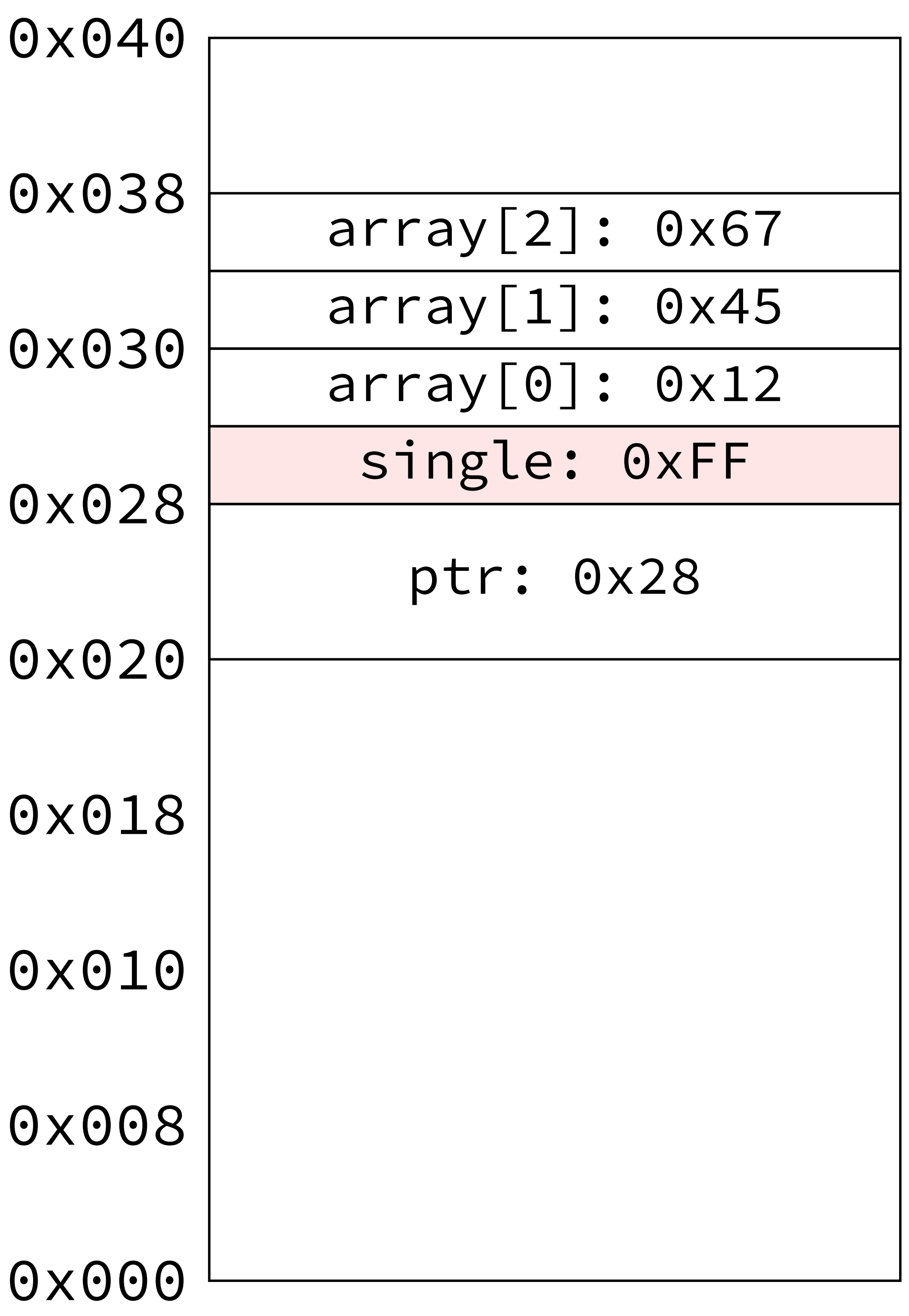

```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;

ptr = &single;
ptr = (int*) 0x28;
```
addr. of single

~~`ptr = 0x28;`~~ compile error (hopefully)

~~`ptr = (int*) single;`~~

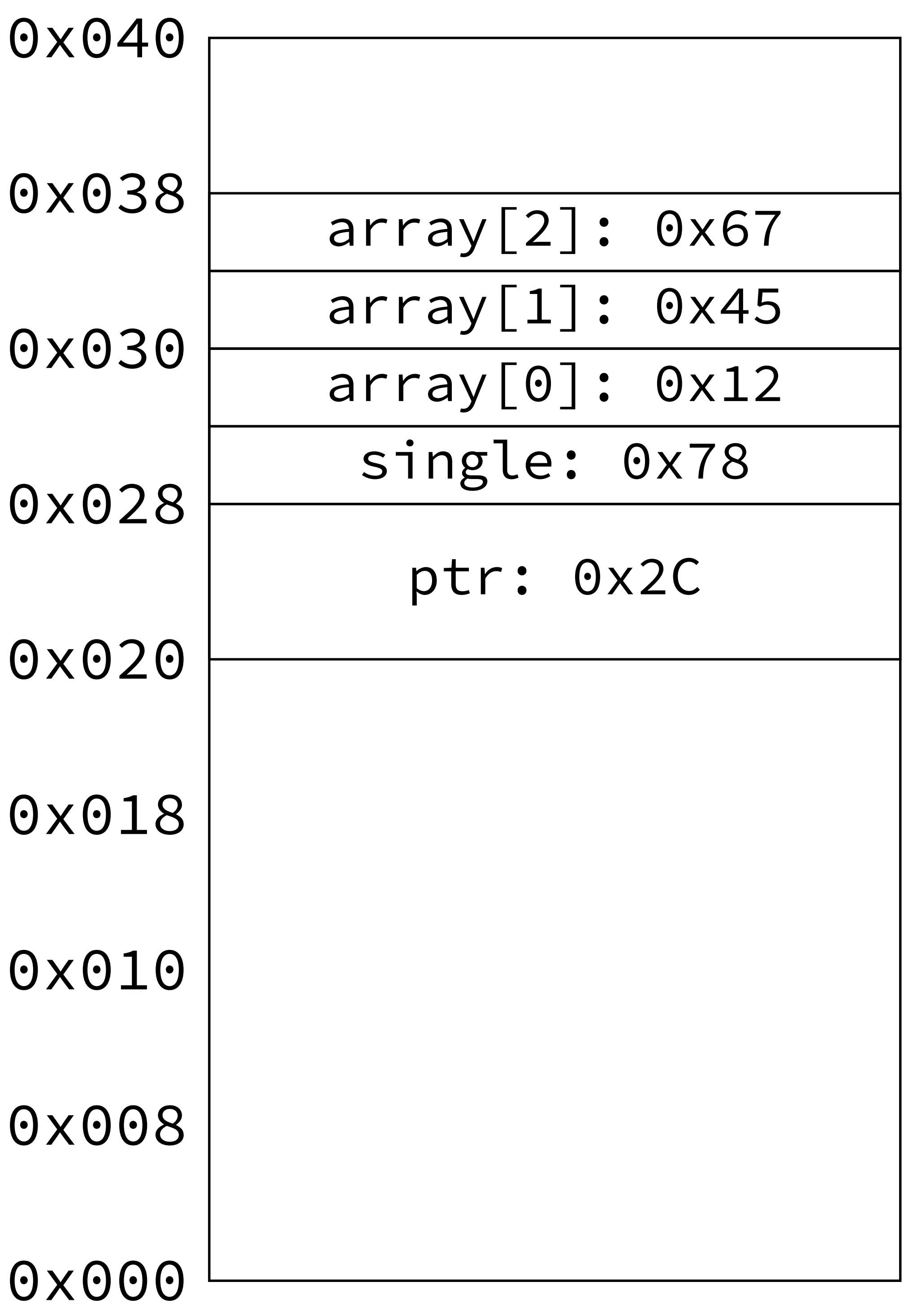
pointer to unknown place

some pointer stuff



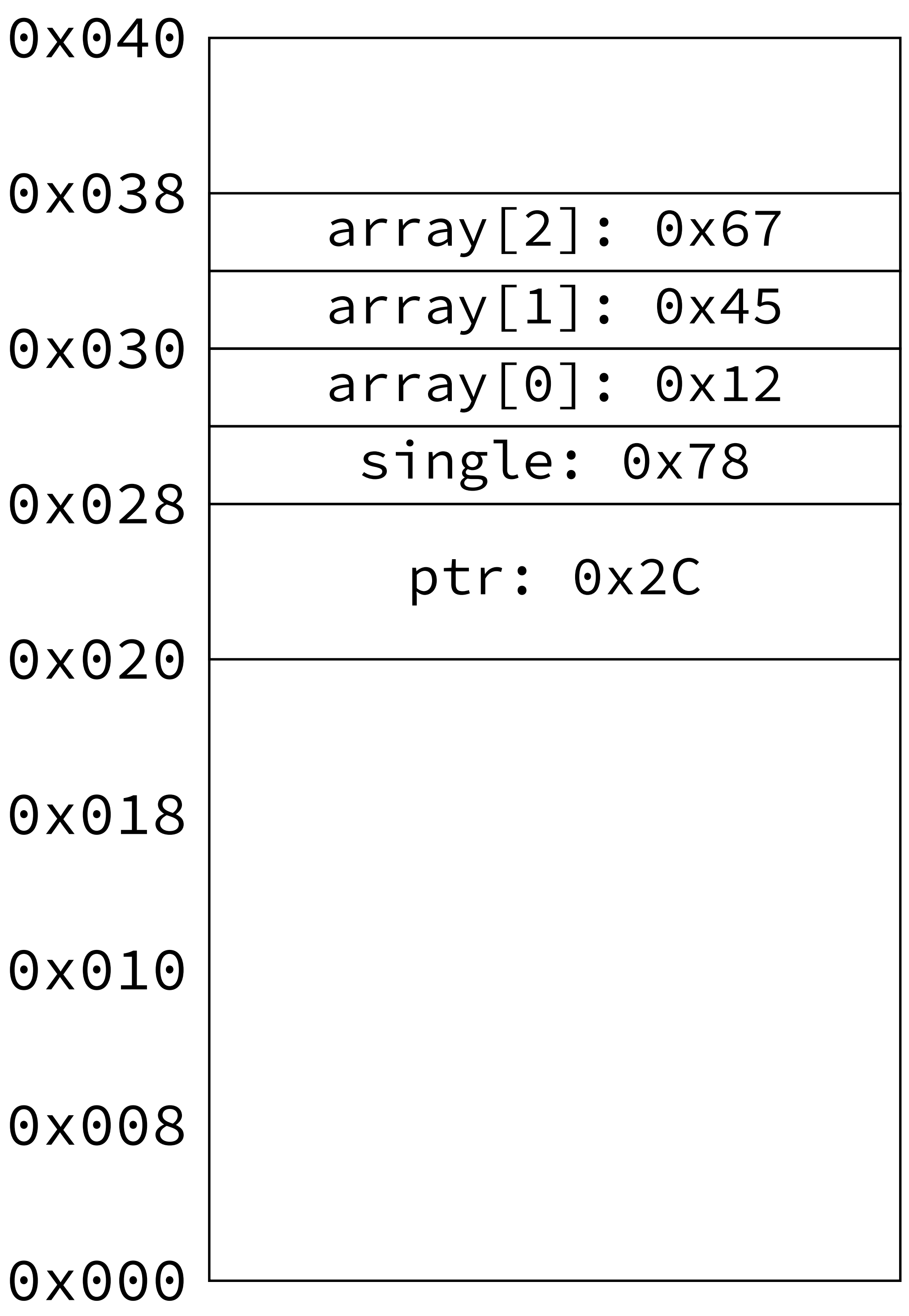

```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;
ptr = &single;

*ptr = 0xFF;
```

some pointer stuff




```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;

ptr = array;
ptr = &array[0];
ptr = (int*) 0x2C;
```

some pointer stuff



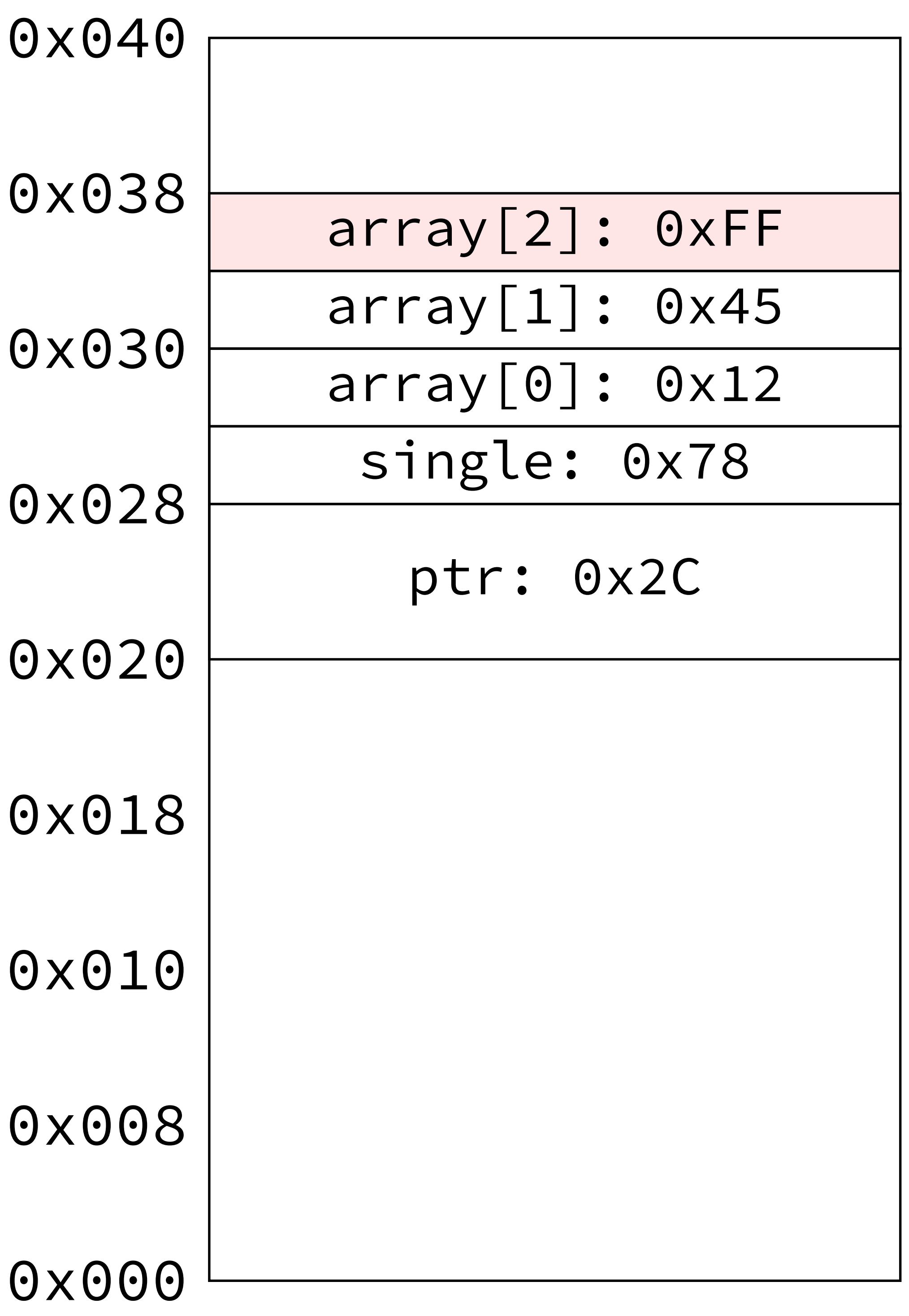

```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;

ptr = array;
ptr = &array[0];
ptr = (int*) 0x2C;
```

~~`ptr = array[0];`~~ compile error*

~~`ptr = (int*) array[0];`~~

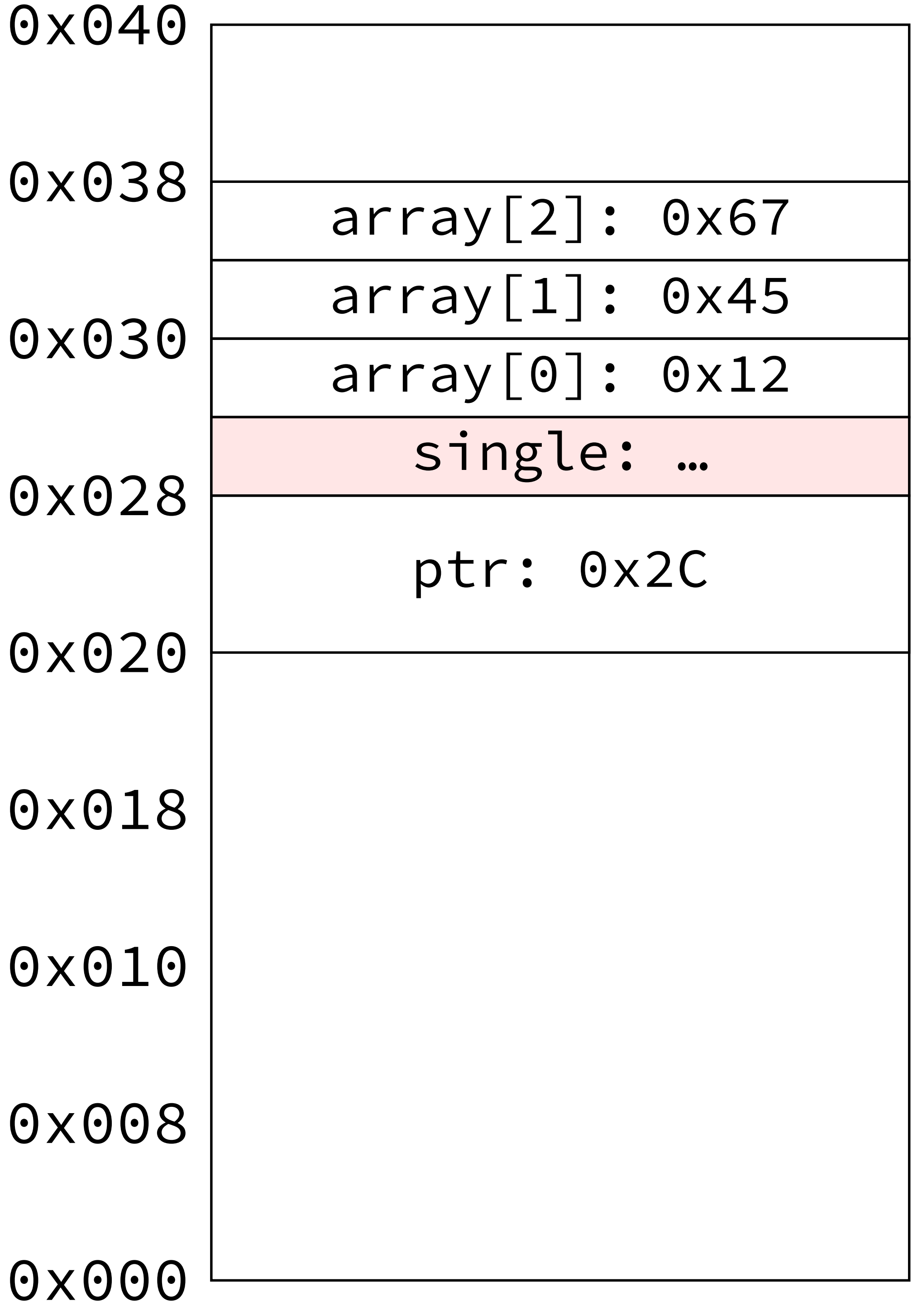
pointer to unknown place

some pointer stuff




```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;
ptr = &array[0];

ptr[2] = 0xFF;
*(ptr + 2) = 0xFF;

int *temp1; temp1 = ptr + 2;
*temp1 = 0xFF;

int *temp2; temp2 = &ptr[2];
*temp2 = 0xFF;
```

some pointer stuff




```
int array[3]={0x12,0x45,0x67};
int single = 0x78;
int *ptr;

void change_arg(int *x) {
      *x = compute_some_value();
}
…
change_arg(&single);
```